

Jinpei Guo

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INFORMATION 1143 Coleman Ave, San Jose, CA 95110 Homepage, GitHub

RESEARCH Low-Level Computer Vision, Diffusion Models, Combinatorial Optimization.
INTERESTS

EDUCATION **Carnegie Mellon University**, Pittsburgh, PA, United States Aug 2024 – Dec 2025
Master, Computer Science
Shanghai Jiao Tong University, Shanghai, China Sep 2020 – Jul 2024
Bachelor, Computer Science

PUBLICATIONS * denotes equal contribution

1. **Jinpei Guo**, Yifei Ji, Zheng Chen, Yufei Wang, Sizhuo Ma, Yong Guo, Yulun Zhang, Jian Wang, “Towards Redundancy Reduction in Diffusion Models for Efficient Video Super-Resolution,” *arXiv*, 2025.
2. **Jinpei Guo**, Yifei Ji, Zheng Chen, Kai Liu, Min Liu, Wang Rao, Wenbo Li, Yong Guo, and Yulun Zhang, “OSCAR: One-Step Diffusion Codec for Image Compression Across Multiple Bit-rates,” *Neural Information Processing Systems (NeurIPS)*, 2025.
3. **Jinpei Guo**, Zheng Chen, Wenbo Li, Yong Guo, and Yulun Zhang, “Compression-Aware One-Step Diffusion Model for JPEG Artifact Removal,” *International Conference on Computer Vision (ICCV)*, 2025.
4. Tingyu Yang, Jue Gong, **Jinpei Guo**, Wenbo Li, Yong Guo, Yulun Zhang, “SODiff: Semantic-Oriented Diffusion Model for JPEG Compression Artifacts Removal,” *Association for the Advancement of Artificial Intelligence (AAAI)*, 2025.
5. Zheng Chen, Mingde Zhou, **Jinpei Guo**, Jiale Yuan, Yifei Ji, Yulun Zhang, “Steering One-Step Diffusion Model with Fidelity-Rich Decoder for Fast Image Compression,” *Association for the Advancement of Artificial Intelligence (AAAI)*, 2025.
6. Yang Li*, **Jinpei Guo***, Runzhong Wang, Hongyuan Zha, and Junchi Yan, “Fast T2T: Optimization Consistency Speeds Up Diffusion-Based Training-to-Testing Solving for Combinatorial Optimization,” *Neural Information Processing Systems (NeurIPS)*, 2024.
7. **Jinpei Guo**, Shaofeng Zhang, Runzhong Wang, Chang Liu, and Junchi Yan, “GMTR: Graph Matching Transformers,” *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2024.
8. Zhaoyu Li, **Jinpei Guo**, Xujie Si, “G4SATBench: Benchmarking and Advancing SAT Solving with Graph Neural Networks,” *Transactions on Machine Learning Research (TMLR)*, 2024.
9. Yang Li, **Jinpei Guo**, Runzhong Wang, and Junchi Yan, “From Distribution Learning in Training to Gradient Search in Testing for Combinatorial Optimization,” *Neural Information Processing Systems (NeurIPS)*, 2023.
10. Zhaoyu Li, **Jinpei Guo**, Yuhe Jiang, and Xujie Si, “Learning Reliable Interpretations with SATNet,” *Neural Information Processing Systems (NeurIPS)*, 2023.

WORK EXPERIENCE	Machine Learning Engineer	Feb 2026 – Present
	ByteDance in San Jose, CA, United States	
	Working on production machine learning systems for large-scale recommendation and advertising, focusing on generative recommendation models, model distillation, semantic ID representations, and efficient Transformer-based architectures for sequential user behavior modeling.	
	Research Assistant	May 2025 – Aug 2025
	Carnegie Mellon University in Pittsburgh, PA, United States	
	Conducted research with Prof. Srinivasa Narasimhan on video super-resolution, focusing on leveraging spatial-temporal information across consecutive frames to improve reconstruction quality and temporal consistency.	
	Machine Learning Engineer Intern	Apr 2024 – Jul 2024
	Ant Group in Shanghai, China	
	Developed an LLM-driven time-series forecasting pipeline integrating a novel “time2text” embedding for sequential data modeling, improving generalization across diverse datasets and enabling unified prediction within a single framework.	
GIFT FUNDINGS AND SCHOLARSHIP	• Snap Inc. Gift Funding, ¥108,000	2025
	• Chiang Chen Overseas Graduate Fellowship (10 recipients nationwide), ¥360,000	2024
	• SenseTime Scholarship (30 recipients nationwide), ¥20,000	2023
	• Chinese National Scholarship, ¥8,000×2	2022, 2023
	• Shanghai Scholarship, ¥8,000	2021
SKILLS	• Computing Skills: Algorithms, Data Structure, Machine Learning.	
	• Programming: Python, C/C++, Matlab, L ^A T _E X.	
	• Programming Frameworks: Pytorch, Scikit-Learn, TensorFlow, Keras.	